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Communication

Editor's Choice

GlycoEnzDB: a database of enzymes involved in human glycosylation

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Abstract

The glycan distribution on cells is governed by the activities of different families of enzymes that are together called "glycoEnzymes." These include ∼400 gene products or 2% of the proteome, that have recently been curated in an ontology called GlycoEnzOnto. With the goal of making this ontology more accessible to the larger biomedical and biotechnology community, we organized a web resource called GlycoEnzDB, presenting this enzyme classification in terms of enzyme function, the pathways that they participate in and their EC numbers. This information is linked to i) Figures from the "Essentials of Glycobiology" textbook, ii) General gene, enzyme and pathway data appearing in external databases, iii) Manual and generative-artificial intelligence (AI) based text describing the function and pathways regulated by these entities, iv) Single-cell expression data across cell lines, normal human cell-types and tissue, and v) CRISPR-knockout/activation/inactivation and Transcription factor activity predictions. Whereas these data are curated for human glycoEnzymes, the knowledge framework may be extended to other species also. The user–friendly web interface is accessible at www.virtualglycome.org/glycoenzdb.

Keywords: glycoinformatics; glycoscience; glycosidase; glycosyltransferase; pathway.

Introduction

Glycosylation is a ubiquitous post-translational modification in mammals that commonly results in the formation of complex carbohydrates or glycans, on diverse protein and lipid scaffolds, mediating numerous biological functions (Varki 2017). The formation of glycan structures is driven by the action of ∼400 genes that are called "glycoGenes" (Neelamegham and Mahal 2016). Each of these glycoGenes is considered to produce corresponding "glycoEnzymes," even though not all molecules strictly display enzymatic activity. These glycoEnzymes, which constitute ∼2% of the expressed proteome, have recently been classified based on an ontology called "GlycoEnzOnto" (Groth et al. 2022). Such semantic literature-based organization of knowledge allows for the description of interactions among these molecules. The curation in GlycoEnzOnto is based on the Gene Ontology convention (Ashburner et al. 2000), where each entry is described in terms of its: i) function or molecular role, ii) pathway or biological process it regulates, and iii) localization or physical location in or outside cells. As glycoEnzymes largely lie in the endoplasmic reticulum or Golgi with a few involved in lysosomal degradation, much of the focus of this manuscript is on "function" and "pathway" relations.

The goal of this manuscript is to introduce a resource called GlycoEnzDB that is freely available as part of the virtualglycome.org website. The effort is distinct from previous work in the field. Notably, the GlycoGene Database (GGDB), which is now part of the GlyCosmos portal (Yamada et al. 2020), curates a limited set of glycosylation genes. The gene set encompassed by GlycoEnzDB is more comprehensive and it is presented as an extension of a well-defined ontology. The Moremen laboratory has curated glycosylation pathways in mammals but these pathways are only associated with limited functional data, primarily focused on recombinant DNA technology and enzyme production (https://glycoenzymes.ccrc.uga.edu/, (Moremen et al. 2012)). Glycopacity (https://glyco.me/) uses gene expression to assess the capacity of individual glycosylation pathways to produce glycans along specified pathways (Schjoldager et al. 2020; Dworkin et al. 2022). In contrast, this paper does not focus on capacity, but rather attempts to link glycobiological knowledge with textbooks in the field, expression data in cell lines and single-cells, and also estimates related to CRISPR sgRNA efficiency and putative transcription factor-glycogene relations.

The focus of GlycoEnzDB is both on new investigators in the field attempting to grasp glycosylation pathways and more experienced scientists looking for well-organized experimental data. One of its goals is to provide an intuitive interface that would be helpful to experimentalists in the biochemistry and biomedical sciences fields. Another goal is to provide a platform for collating enzyme-centric data related to glycosylation

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pathways. As a starting point, a limited set of experimental data are curated in the current GlycoEnzDB framework, and we expect to expand on this in the future based on our ongoing research.

Results and discussion

The glycoEnzymes are an important starting point for studies of glycobiology, as they define the reaction pathways used to produce glycans. GlycoEnzDB curates 403 human glycoEnzymes based on their function (Table S1) and the pathways (Table S2) they participate in. From the "function" perspective, the glycoEnzymes are grouped into glycosyltransferases (sialyltransferases, fucosyltransferases, etc.), other transferases (e.g. sulfotransferases), modifying enzymes (e.g. epimerases and kinases), glycosidases, molecular transporters, and other regulators. Additionally, the glycoEnzymes are also classified according to their "pathways" in terms of their roles in glycoconjugate biosynthesis, degradation, donor synthesis, and transport. These biosynthesis pathways include the steps involved in: (i) "Core" biosynthesis that results in the attachment of the first monosaccharide(s) or oligosaccharide to the protein/lipid that initiates the biosynthesis of specific glycoconjugate types. In this regard, the core pathways initiate the biosynthesis of glycolipids (glycosphingolipid and GPI pathways), glycosaminoglycans (hyaluronic acid, chondroitin/dermatan sulfate, heparan sulfate), O-linked glycans (O-GalNAc type, O-GlcNAc and additional rarer modifications), C-Mannosyl linkages at Trp, and N-linked glycans commonly at the Asn-X-Ser/Thr sequon; (ii) "Elongation and branching" reactions that extend the original glycan via the extension of chondroitin/dermatan, heparan, and keratan sulfate chains, LacNAc (N-acetyl lactosamine), LacdiNAc (GalNAc(β1–4)GlcNAcβ) and I-branching (Gal(1–4)[GlcNAc(β1–6)]Galβ); and (iii) "Termination or capping" processes that often involve sulfation, fucosylation and sialylation reactions.

Many of the above functions and pathways are linked to 24 figures that are adapted from the Essentials of Glycobiology (EoG) textbook that capture the human glycosylation biosynthesis process (Varki et al. 2022). Figure 1A presents a screenshot of the front page of the database. Clicking on the "Function," "Pathways" or "EC Numbers" buttons displays a list of genes having corresponding function, pathway or EC numbers based on GlycoEnzOnto. A relevant figure from the EoG text is presented in many cases. The enzyme names in these figures appear in red, and clicking on any name will take the user to more detailed information pages described below. Additionally, if enzymes are searched based on their HGNC gene name, this also opens the details page for that enzyme along with corresponding EoG figure(s), when available. To better explain the enzyme activity, we have manually curated text related to individual enzyme function. This is primarily collated from the EoG textbook (Varki et al. 2022) and the "Handbook of glycosyltransferases and related genes" (Taniguchi et al. 2014). The EoG text was also used to train a custom GPT (Generative pre-trained transformer model) called "Dr. Glyco-GPT." This new GPT can be queried with glycosylation related questions. For GlycoEnzDB, the GPT was asked to write a short essay about each of the curated enzymes, with respect to their: i) function, ii) reaction pathways, ii) location, and iv) disease relation. The output is presented in the database, with only limited manual curation. Dr. Glyco-GPT can also be used to answer additional user queries, but one must caution that the use of large-language models (LLMs) can be error-prone and assertions need to be validated manually against published literature. Additionally, DrawGlycan-SNFG is used to draw reaction schemes describing the catalytic activity of the individual glycoEnzymes (Cheng et al. 2017; Cheng et al. 2019). This uses the Symbol Nomenclature for Glycans (SNFG) nomenclature for rendering pictorial schematics of complex carbohydrates (Varki et al. 2015; Neelamegham et al. 2019).

Besides manual curation, GlycoEnzDB also integrates information for each enzyme using external databases listed in Table 1. These include resources that provide community-consensus HGNC gene names (Seal et al. 2023), and databases related to nucleic acids (NCBI: (Haft et al. 2024)), proteins (UniProt: (UniProt 2023)), glycans (GlyGen: (York et al. 2020)), diseases (OMIM: (Amberger and Hamosh 2017)), enzymes (CAZy: (Drula et al. 2022), IUBMB: (McDonald et al. 2009), Brenda: (Chang et al. 2021)), reaction rules (Rhea: (Bansal et al. 2022)) and pathway maps (KEGG: (Hashimoto et al. 2006), Reactome: (Milacic et al. 2024)) (Fig. 1B). Additional wet-lab data and computational predictions are provided to connect textbook knowledge with experiments. Currently, this is done by scraping glycogene expression data for various cancer cell lines from DepMap (Ghandi et al. 2019), and single-cell glycogene data from Tabula Sapiens (Tabula Sapiens et al. 2022). In this regard, DepMap includes bulk sequencing data for ~2000 cell lines including many of the cells that are part of the NCI-60 collection. The scRNA-seq (single-cell RNA-seq) data from Tabula Sapiens comprises measurements from 483,152 human cells from 15 donors that are organized into 24 tissues and over 400 cell types. We also provide CRISPR knockout (CRISPR-ko), activation (CRISPR-a) and inactivation (CRISPR-i) sequences from the CRISPick resource (Doench et al. 2016). Finally, the database provides computational predictions of TFs regulating glycosylation by examining mutual information of scRNA-seq from the Tabula Sapiens dataset for every TF-glycoGene pair from the TF-links database (Chrysinas et al. 2024). Figure 2 provides webpage vignettes for a representative enzyme.

Although GlycoEnzDB is curated for human glycoenzymes, many of the same or similar pathways occur in other mammals, with a few exceptions. Among these is Cytidine monophospho-N-acetylneuraminic acid hydroxylase (CMAH), which catalyzes the conversion of Neu5Ac (N-acetylneuraminic acid) to Neu5Gc (N-glycolylneuraminic acid). CMAH is present in Old World primates and many other deuterostome species but is inactive in *Homo sapiens* and has been independently lost in several other taxa, including the common ancestor of all birds. GGTA1 (α3Gal-T), which transfers Gal from UDP-Gal to the nonreducing terminal β-Gal of glycoconjugates to form alpha-Gal (Galα1-3Gal), is found in most non-primate mammals but is absent in Old World primates, including humans, due to gene inactivation. Finally, GBGT1, which provides α(1–3)GalNAc activity necessary to generate the Forssman glycolipid antigen, is inactive in most humans, although functional alleles persist in a minority of individuals. Scripts are available on the GlycoEnzDB GitHub repository to extend the database to non-human species.

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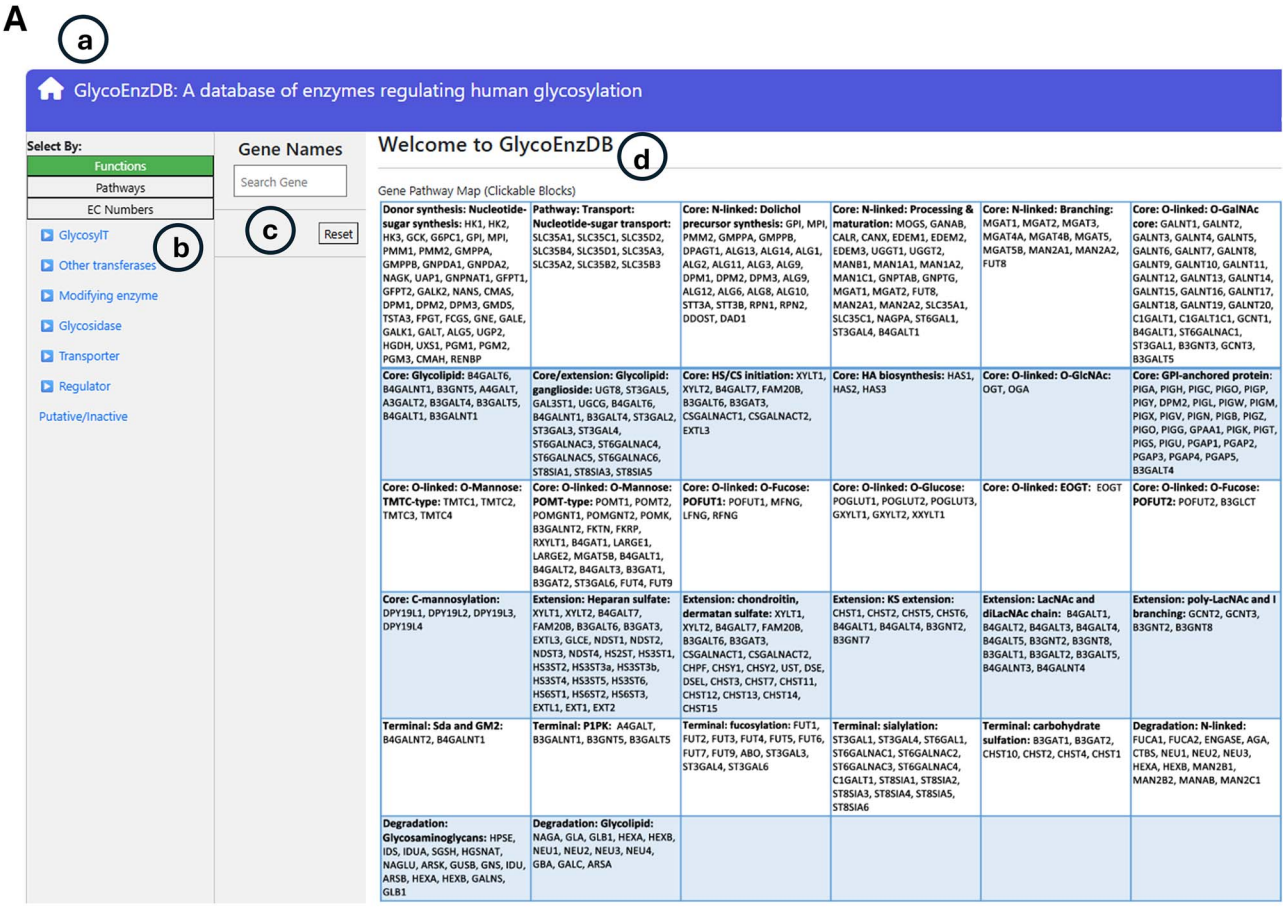


Figure 1. GlycoEnzDB: A) Landing page for web resource. Icons highlight: a) home link to return to this page; b) pull down menus to parse GlycoGenes/enzymes based on function, pathway or EC number; c) search field based on gene HGNC name; d) 32 mammalian glycosylation pathways curated from the essentials of glycobiology textbook. B) Other resources and links available at GlycoEnzDB database.

The current version of GlycoEnzDB is focused on a limited set of experimental data. We anticipate that this will grow in the coming years with large-scale characterization of microRNA, epigenetic factors, and regulatory elements controlling glycosylation. We will endeavor to add these insights as they emerge. The enzyme rules and pathway maps provided by the resource may also facilitate the automated development of custom reaction pathway maps for model building (Neelamegham and Liu 2011; Liu and Neelamegham 2015; Huang et al. 2021). Such integration of knowledge is important for establishing the highly non-linear relationship between gene expression and glycan structures, and for better understanding

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Table 1. Sources of data curated at GlycoEnzDB.

Category	Description	Web link	Citation
Overall organization:			
Glycosylation enzyme ontology	GlycoEnzOnto: ontology for ~400 glycoGenes	https://github.com/neel-lab/GlycoEnzOnto	(Groth et al. 2022)
General Information:			
Figures	Essentials of Glycobiology	https://www.ncbi.nlm.nih.gov/books/NBK579918/	(Varki et al. 2022)
Gene name	HUGO Gene Nomenclature Committee (HGNC)	https://www.genenames.org/	(Seal et al. 2023)
Gene: Reference sequence	National Center for Biotechnology Information (NCBI)	https://www.ncbi.nlm.nih.gov/gene	(Haft et al. 2024)
Protein data, synonyms, short/alternate names	UniProt: Universal Protein Knowledgebase	https://www.uniprot.org/	(UniProt 2023)
Carbohydrate enzyme data	Carbohydrate-Active enZYmes Database	http://www.cazy.org/	(Drula et al. 2022)
Glycosylation specific data	Glygen: Computational and Informatics Resources for Glycosciences	https://glygen.org/	(York et al. 2020)
Description:			
Manual notes	Manual curation from textbooks, journal articles and handbooks	http://virtualglycome.org/glycoenzdb	(Taniguchi et al. 2014, Varki et al. 2022)
Generative AI	ChatGPT trained using glycobiology textbook	https://chatgpt.com/g/g-oVvZ8xe4a-dr-glyco	
Reaction schematic	Manual curation of enzyme action, with figures rendered using DrawGlycan-SNFG software	http://virtualglycome.org/glycoenzdb	(Cheng et al. 2017, Cheng et al. 2019)
Reaction and disease links:			
EC number	International Union of Biochemistry and Molecular Biology (IUBMB)	https://www.enzyme-database.org/	(McDonald et al. 2009)
Catalytic data	BRENDA: Enzyme functional data	https://www.brenda-enzymes.org/	(Chang et al. 2021)
Disease association	OMIN: Online Mendelian Inheritance in Man	https://www.omim.org/	(Amberger and Hamosh 2017)
Pathway map	KEGG: Kyoto Encyclopedia of Genes and Genomes	https://www.genome.jp/kegg/glycan/	(Hashimoto et al. 2006)
Curated chemical/transport reactions	Expert-curated reactions of biological interest	https://www.rhea-db.org/	(Bansal et al. 2022)
Reaction data	peer-reviewed pathway database.	https://reactome.org/	(Milacic et al. 2024)
Functional data:			
Expression in cell lines	CCLE (Cancer cell line encyclopedia)	https://depmap.org/portal/	(Ghandi et al. 2019)
Expression in single human cells	The tabula sapiens consortium	https://tabula-sapiens.sf.czbiohub.org/	(Tabula Sapiens et al. 2022)
CRISPR data	CRISPick: Rank and picks candidate CRISPRko/a/i	https://portals.broadinstitute.org/gppx/crispick/public	(Doench et al. 2016)
Glycogene-TF links	Computational prediction of transcriptional factors (TFs) regulating human glycogenes and glycopathways	http://vgdev.cedar.buffalo.edu/glycotf/	(Chrysinas et al. 2024)

how glycoenzymes competing for the same substrate quantitatively regulate the glycan distribution on cells.

Materials and methods

Sources of data

The GlycoEnzDB is built on the Django python framework using the PostgreSQL relational database management system. This is an open-source resource. Source data are provided at https://github.com/neel-lab/webGlycoEnzDB, and they come from original resources listed in Table 1. This GitHub repository also contains code used to generate the website, data tables and accompanying figures.

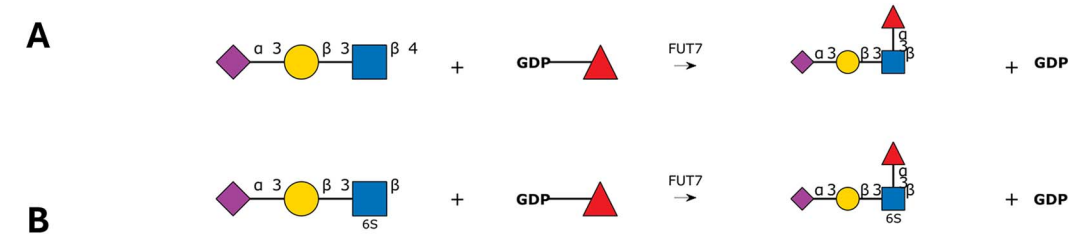
General information and enzyme description

General information about each enzyme including the gene name and links to external databases related to enzymes and pathways were gleaned from the UniProt server (UniProt 2023). Notes related to enzymes were mostly manually curated from textbooks in the field. Reaction figures were rendered using the DrawGlycan-SNFG software (Cheng et al. 2017; Cheng et al. 2019), based on recommended SNFG standards (Varki et al. 2015; Neelamegham et al. 2019).

Dr.Glyco-GPT

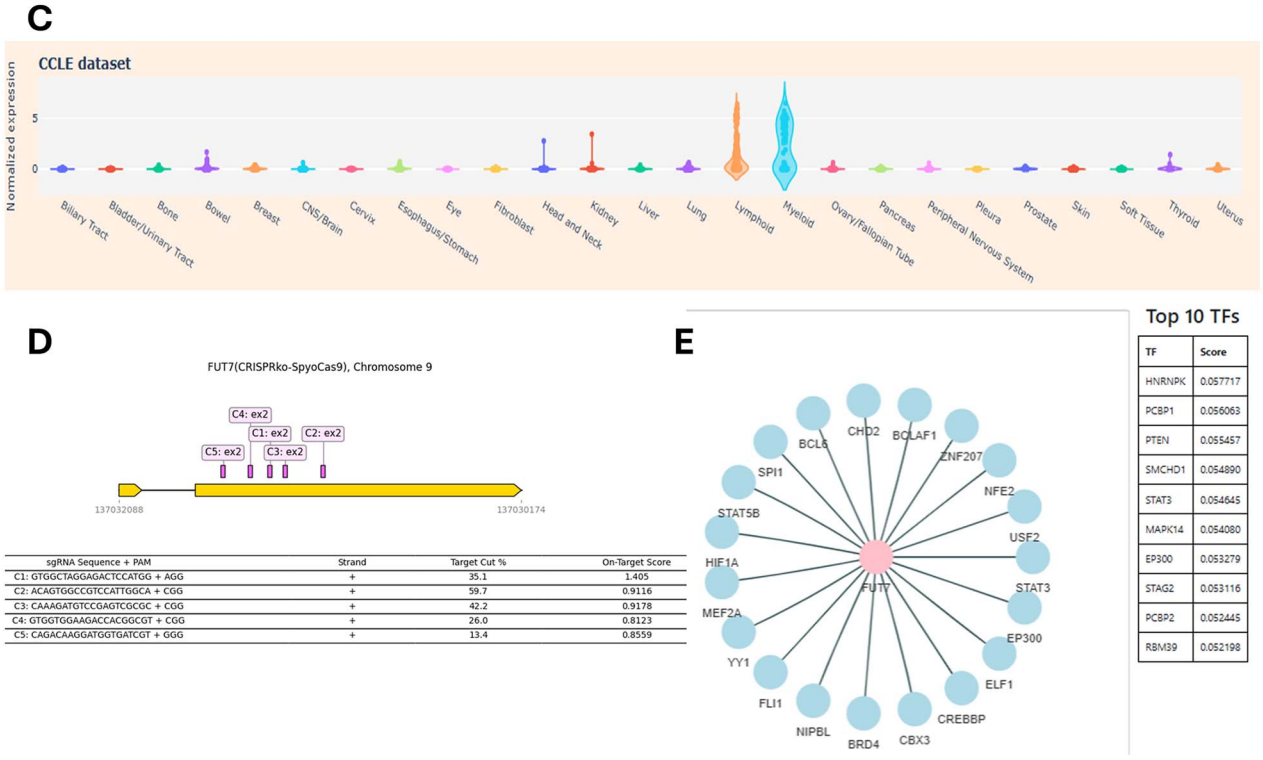
A custom GPT was generated using tools at ChatGPT.com. Knowledge from the EoG (Varki et al. 2022) was input used for training, with permission from the textbook editors.

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B

```
### Function Human FUT7 (fucosyltransferase 7) is a crucial enzyme that catalyzes the addition of fucose in an α1,3-linkage
to N-acetylglucosamine residues on glycoproteins and glycolipids. This activity is specifically involved in the synthesis of
the sialyl Lewis X (sLex) antigen, a glycan structure that plays a significant role in mediating leukocyte adhesion to
endothelial cells during inflammation...
### Reaction Pathways FUT7 operates in the fucosylation pathway, where it transfers fucose from GDP-fucose to sialylated
type 2 lactosamine chains, forming the sLex antigen. This reaction is critical for creating ligands for selectins (E-, P-,
and L-selectins), which mediate the rolling of leukocytes on the endothelial surface during the initial steps of
inflammation and immune response...
### Location FUT7 is localized in the Golgi apparatus, where it carries out its function in the glycosylation process. The
Golgi apparatus is the site where glycoproteins and glycolipids are processed and modified before being sent to their final
destinations...
### Diseases Mutations or altered expression of FUT7 can lead to deficiencies in sLex expression, impairing leukocyte
adhesion and migration. Such deficiencies can result in weakened immune responses and are associated with conditions like
leukocyte adhesion deficiency type II (LAD II), which leads to recurrent bacterial infections and other immune-related
issues...
```



sgRNA Sequence + PAM	Strand	Target Cut %	On-Target Score
C1: GTGGCTAGGAGACTCCATGG + AGG	+	35.1	1.405
C2: ACAGTGGCCGTCCATTGGCA + CGG	+	59.7	0.9116
C3: CAAAGATGTCCGAGTCGCGC + CGG	+	42.2	0.9178
C4: GTGGTGGAAGACCACGGCGT + CGG	+	26.0	0.8123
C5: CAGACAAGGATGGTGATCGT + GGG	+	13.4	0.8559

TF	Score
HNRNPK	0.057717
PCBP1	0.056063
PTEN	0.055457
SMCHD1	0.054890
STAT3	0.054645
MAPK14	0.054080
EP300	0.053279
STAG2	0.053116
PCBP2	0.052445
RBM39	0.052198

Figure 2. Figure vignettes: Sample data for FUT7 showing: A) Catalytic reaction. B) Sample Dr.Glyco-GPT text. C) Cell line transcript expression data; D) CRISPR knockout figure; E) Putative link between glycogenes and transcription factors.

The generative resource was then prompted to write a 100–350 word essay about each of the enzymes with respect to their: i) function, ii) reaction pathways, ii) location, and iv) diseases. Note that, while language models such as ChatGPT are powerful tools for information synthesis, the output can be incorrect or outdated. Users, thus, need to consider the output in context of peer-reviewed literature.

Cancer cell expression data (CCLE)

TPM values for all DepMap cell types was downloaded in April 2023 using the "OmicsExpressionProteinCodingGenesTPMLogp1.csv" file. Non-glycogenes were removed and all DepMap cells were preserved. These data are plotted as violin plots. Values are inferred from RNA-seq data using the RSEM tool and are reported after log2 transformation, i.e. log2(TPM + 1).

Single-cell expression data

10X genomics based scRNA-seq data (456,101 cells) were from the Tabula Sapiens (TS) project (Tabula Sapiens et al. 2022). Data were processed as described previously (Chrysinas et al. 2024). Briefly, the UMI (unique molecular identifier) counts for each cell were normalized to 10,000 and log transformed (log1p), to obtain relative RNA abundance data. These are plotted for the enzymes in the GlycoEnzOnto (Groth et al. 2022). Plots are generated to quantify transcript

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abundance in individual tissue types and also cell types or "compartments" in that tissue.

CRISPR data

CRISPR knockout, activation and inactivation data were curated from the CRISPick resource using the human GRCh38 reference genome, and *S. pyrogenes* containing the NGG PAM sequence. Five sgRNA were chosen for each glycogene based on a previously published algorithm (Doench et al. 2016). Figures presented at the GlycoEnzDB resource were generated using the DNA_features_viewer package. They present sgRNA target location and "on-target scores" which quantify the theoretical efficacy of each of the sgRNA.

Glycogene-transcription factor relation

TF-glycogene pairs are curated from the TFLink database, and this was previously compiled from numerous databases that relied on different evidence(s) of TF binding to regulatory elements on genes (Liska et al. 2022). Mutual information (MI) was used to provide additional evidence for TF-glycogene interactions based on single-cell expression data from the Tabula Sapiens. Here, MI is calculated according to:

$$I(TF, G) = \sum_{t_i} \sum_{g_j} P_{TF,G}(tf, g) \log \frac{P_{TF,G}(tf, g)}{P_{TF}(tf) P_G(g)} \quad (1)$$

where $P_{TF,G}$denotes the joint probability distribution of single-cell expression of a transcription factor *TF* and a glycogene *G*, and P_{TF} and P_G denote the marginal probability distribution of *TF* and *G*, respectively. We evaluated $I(TF, G)$using the scaled UMI counts from all 10X cells in the TS dataset. More details are available elsewhere (Chrysinas et al. 2024).

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Author contributions

Y.Z., V.G., E.S., S.S., S.V., S.N.: Methodology, Investigation; S.S.: Resources, supervision; R.G: Methodology, investigation, supervision; S.N.: Conceptualization, supervision, methodology, investigation, writing-original draft; All authors: Writing – review & editing.

Supplementary material

Supplementary material is available at *Glycobiology Journal* online.

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Conflicts of interest

The authors declare no financial or non-financial conflicts of interest.

Data availability statement

All data, instructions and source code used to generate the web interface are available at: https://github.com/neel-lab/webGlycoEnzDB. GlycoEnzDB is deployed at: www.virtualglycome.org/glycoenzdb.

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